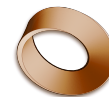


Name: _____

Class: _____

Mr. Rawson • GCSE Computer Science



Year 10 2026

Data Representation

Number Bases & Units of Information

OCR J277 • 1.2.3 Units • 1.2.4 Data Storage — Numbers

01000011 01010011

0x43 0x53 67 83

In this booklet: bits, nibbles and bytes
kB • MB • GB • TB • PB
denary, binary and hexadecimal
converting between all three bases

Why must computers use binary?

A computer is built from millions of tiny electrical switches. A switch has no in-between: it is either **on** or **off**. We label on as **1** and off as **0**, and every one of these single on/off values is called a **bit** — short for *binary digit*.

Two states are easy to tell apart, even when a signal is weak or noisy, which is why binary is so reliable. With ten different voltage levels (one per denary digit) a slightly faded signal could easily be misread as its neighbour; with two levels it almost never is.

On its own, one bit says very little. Joined together, bits form a **bit pattern**, and bit patterns can represent *anything*: numbers, text, images, sound and program instructions are all stored the same way — as patterns of 1s and 0s.

Key concept

Everything in a computer — this booklet included — is stored as patterns of 1s and 0s. Data must be converted into binary before a computer can process it.

What you will learn

- why data must be stored in a binary format
- the units of data storage: bit, nibble, byte, kB, MB, GB, TB, PB
- how to calculate data capacity and storage requirements
- the three number bases: **denary** (base 10), **binary** (base 2) and **hexadecimal** (base 16)
- how to convert between all three, in every direction

Exam limits

In the exam the largest values you will be asked to convert are: denary **255** • binary **1111 1111** • hexadecimal **FF**. *No calculators allowed* — every method in this booklet works on paper.

Warm-up. Which of these could be binary numbers? Circle them.

1011 2001 111111 10A1 0000 1234

Binary: 1011, 111111, 0000 — a binary number may only contain the digits 0 and 1.

Exam question [1 marks]

OCR J277/01 · June 2023 · Q1(a)

Tick (✓) **one** box to identify the statement about binary that is true.

- | | |
|-------------------------------------|--|
| ☐ | Binary digits can only be the values 0, 1 and 2 |
| ☐ | The left-most bit of a binary integer has the smallest value |
| ☒ | Binary is used because computers are made of switches that can only be on or off |
| ☐ | The smallest whole number that can be stored in 8 bits is the number 1 |

Bits, nibbles and bytes

A **bit** is the smallest unit of data a computer can use — a single 1 or 0. A **byte** is a group of **8 bits**, and it is the smallest bit pattern normally used to represent data. Half a byte — **4 bits** — is called a **nibble**.

$\underbrace{1\ 0\ 1\ 0}_{\text{nibble}}\ \underbrace{0\ 0\ 1\ 0}_{\text{nibble}} \quad 1 \text{ byte} = 8 \text{ bits} = 2 \text{ nibbles}$

Key facts

- Abbreviations: **b** = bit, **B** = byte. Case matters!
- 1 nibble gives **16** possible values — 0 to 15
- 1 byte gives **256** possible values — 0 to 255

Worked example

8 b = 8 bits = 1 B = 1 byte 3 B = 3 bytes = 24 bits 5 b = 5 bits

Practice 3.1 — How many *bits* are there in:

a) 2 bytes 16 b) 3 bytes 24 c) 7 bytes 56 d) 10 bytes 80

Practice 3.2 — How many *bytes* are there in:

a) 32 bits 4 b) 72 bits 9 c) 96 bits 12 d) 160 bits 20

Practice 3.3 — How many *nibbles* are there in 3 bytes? 6

Practice 3.4 — A byte holds 256 different values but a nibble only 16. Why does adding 4 more bits multiply the number of values by 16?

Each extra bit doubles the number of possible patterns. Four extra bits double it four times: $2 \times 2 \times 2 \times 2 = 16$. So $16 \times 16 = 256$.

Exam question [1 marks]

OCR J277/01 · June 2025 · Q1(b)(iii)

State the minimum number of bits that will be needed to represent 240 different colours.

8

Quantities of bytes

Files are far bigger than single bytes, so we measure them with the International System of Units, known as **SI** units. Each step up the ladder multiplies by 1,000:

1 kilo byte =	1 kB =	1,000 B =	1,000 bytes =	10^3 bytes
1 mega byte =	1 MB =	1,000 kB =	1,000,000 bytes =	10^6 bytes
1 giga byte =	1 GB =	1,000 MB =	1,000,000,000 bytes =	10^9 bytes
1 tera byte =	1 TB =	1,000 GB =	1,000,000,000,000 bytes =	10^{12} bytes
1 pet a-byte =	1 PB =	1,000 TB =	1,000,000,000,000,000 bytes =	10^{15} bytes

Maths Link – kilos

The prefixes are the same ones you already use in maths and science:

1 kilo gram =	1 kg =	1,000 grams
1 kilo metre =	1 km =	1,000 metres

Practice 4.1 — Convert the following:

- | | |
|------------------------------|------------------------------------|
| a) 8 kB = <u>8,000</u> bytes | b) 23 MB = <u>23,000,000</u> bytes |
| c) 2 MB = <u>2,000</u> kB | d) 8 GB = <u>8,000,000</u> kB |
| e) 42 TB = <u>42,000</u> GB | f) 0.5 TB = <u>500</u> GB |

Practice 4.2 — Write these units in order, smallest to largest:

MB nibble GB bit PB byte kB TB

bit, nibble, byte, kB, MB, GB, TB, PB

Practice 4.3 — A 64 GB memory card stores photos of 4 MB each. How many photos will it hold? Show your working.

64 GB = 64,000 MB. $64,000 \div 4 = 16,000$ photos.

Exam question [4 marks]

OCR J277/01 · June 2022 · Q1(a)

Tick (✓) one box in each row to identify the binary unit equivalent of each of the given file sizes.

File size	2 megabytes	2 petabytes	2 kilobytes	2 bytes	2 gigabytes
2000 bytes			✓		
2000 terabytes		✓			
16 bits				✓	
4 nibbles				✓	

Denary — base 10

A **number base** says how many different digits a number system uses. **Denary** (base 10, also called decimal) is the system you have used all your life. It has ten digits:

0 1 2 3 4 5 6 7 8 9

Once we run out of digits we add a new column to the left, and each column is worth **ten times** the one before — units, tens, hundreds, thousands:

Thousands	Hundreds	Tens	Units
10^3	10^2	10^1	10^0
1000	100	10	1

Worked example

Take **3,512**. Put each digit under its column heading and add up what each is worth:

Thousands	Hundreds	Tens	Units
3	5	1	2

$3,000 + 500 + 10 + 2 = 3,512$

You never think about this — that is the point. Binary and hexadecimal work by *exactly the same rule*; only the size of the columns changes.

Practice 5.1 — Break these denary numbers into their column values, as in the worked example:

	Thousands	Hundreds	Tens	Units
a) 4,807	4	8	0	7
b) 926	0	9	2	6
c) 5,237	5	2	3	7

a) $4,807 = \underline{\hspace{2cm} \mathbf{4,000 + 800 + 0 + 7} \hspace{2cm}}$

b) $926 = \underline{\hspace{2cm} \mathbf{900 + 20 + 6} \hspace{2cm}}$

Binary — base 2

Binary (base 2) is the same idea with only **two** digits:

0 1

Because there are only two digits, each column is worth **twice** the one before — units, twos, fours, eights:

Eights	Fours	Twos	Units
2^3	2^2	2^1	2^0
8	4	2	1

Worked example

Take the binary number **1011**. Put each bit under its column and add up the columns that contain a 1:

Eights	Fours	Twos	Units
1	0	1	1

$8 + 0 + 2 + 1 = 11$ so 1011 in binary = **11** in denary.

Maths Link – powers of 2

The column values are the powers of 2: $2^0 = 1$, $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 16$, $2^5 = 32$, $2^6 = 64$, $2^7 = 128$. Doubling each time — learn this line and you can rebuild any conversion table from memory.

Practice 6.1 — Convert these nibbles to denary using the 8–4–2–1 columns:

a) 1001 = **9** b) 1111 = **15** c) 0110 = **6** d) 1100 = **12**

Practice 6.2 — What is the largest denary value a nibble can hold? **15** And the smallest? **0**

Binary → denary

For a full byte we extend the column headings to eight places. This is the **binary line** — write it out as soon as your exam starts:

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

The method: write the binary number under the line, one bit per column, then add up the value of every column that holds a 1.

Worked example

Convert 1101 1011 to denary.

128	64	32	16	8	4	2	1
1	1	0	1	1	0	1	1

$128 + 64 + 16 + 8 + 2 + 1 = \mathbf{219}$

Practice 7.1 — Convert to denary. Use the empty rows to write the binary line and the bits underneath.

Binary	Working	Denary
a) 1110		<u>14</u>
	$8+4+2$	
b) 101010		<u>42</u>
	$32+8+2$	
c) 1100 1100		<u>204</u>
	$128+64+8+4$	
d) 0101 0101		<u>85</u>
	$64+16+4+1$	
e) 1000 0001		<u>129</u>
	$128+1$	
f) 1111 1111		<u>255</u>
	$128+64+32+16+8+4+2+1$	

Practice 7.2 — Explain why 1111 1111 is the largest number the exam will ever ask you to convert. **It is the largest value of one byte (8 bits) — every column of the binary line is switched on: 255.**

Denary → binary

To go the other way, **repeatedly divide by 2** and keep the remainders. Each remainder is one bit of the answer.

Worked example

Convert **156** to binary.

$$\begin{array}{rcl} 156 \div 2 = 78 & \text{remainder} & \mathbf{0} \\ 78 \div 2 = 39 & \text{remainder} & \mathbf{0} \\ 39 \div 2 = 19 & \text{remainder} & \mathbf{1} \\ 19 \div 2 = 9 & \text{remainder} & \mathbf{1} \\ 9 \div 2 = 4 & \text{remainder} & \mathbf{1} \\ 4 \div 2 = 2 & \text{remainder} & \mathbf{0} \\ 2 \div 2 = 1 & \text{remainder} & \mathbf{0} \\ 1 \div 2 = 0 & \text{remainder} & \mathbf{1} \end{array}$$

Stop when the answer reaches 0, then read the remainders **from bottom to top**: $156 = 1001\ 1100$

Check your answer

Convert back the other way: $128 + 16 + 8 + 4 = 156$. ✓ A conversion you can check both ways is a conversion you cannot get wrong.

Practice 8.1 — Convert to binary. Show the division ladder.

a) 21

$$\begin{array}{l} 21 \div 2 = 10 \text{ r } 1; 10 \div 2 = 5 \text{ r } 0; 5 \div 2 = 2 \text{ r } 1; \\ 2 \div 2 = 1 \text{ r } 0; 1 \div 2 = 0 \text{ r } 1 \rightarrow 10101 \end{array}$$

b) 45

$$\begin{array}{l} 45 \div 2 = 22 \text{ r } 1; 22 \div 2 = 11 \text{ r } 0; 11 \div 2 = 5 \text{ r } 1; \\ 5 \div 2 = 2 \text{ r } 1; 2 \div 2 = 1 \text{ r } 0; 1 \div 2 = 0 \text{ r } 1 \rightarrow 101101 \end{array}$$

c) 164

$$\begin{array}{l} 164 \div 2 = 82 \text{ r } 0; 82 \div 2 = 41 \text{ r } 0; 41 \div 2 = 20 \text{ r } 1; \\ 20 \div 2 = 10 \text{ r } 0; 10 \div 2 = 5 \text{ r } 0; 5 \div 2 = 2 \text{ r } 1; \\ 2 \div 2 = 1 \text{ r } 0; 1 \div 2 = 0 \text{ r } 1 \rightarrow 1010\ 0100 \end{array}$$

d) 230

$$\begin{array}{l} 230 \div 2 = 115 \text{ r } 0; 115 \div 2 = 57 \text{ r } 1; 57 \div 2 = 28 \\ \text{r } 1; 28 \div 2 = 14 \text{ r } 0; 14 \div 2 = 7 \text{ r } 0; 7 \div 2 = 3 \text{ r } 1; \\ 3 \div 2 = 1 \text{ r } 1; 1 \div 2 = 0 \text{ r } 1 \rightarrow 1110\ 0110 \end{array}$$

Exam question [3 marks]

OCR J277/01 · June 2024 · Q1(a)

Complete the table by converting the 8-bit binary number into denary and the denary number into 8-bit binary.

8-bit Binary	Denary
11110000	<u>240</u>
<u>01101001</u>	105
00011110	<u>30</u>

Hexadecimal — base 16

Hexadecimal (base 16, or just *hex*) uses **sixteen** digits. After 9 we run out of ordinary digits, so the letters A to F stand for the values ten to fifteen:

0 1 2 3 4 5 6 7 8 9 A B C D E F

Why does hex exist? Long bit patterns are hard for humans to read, remember and copy without mistakes. Hexadecimal is a compact, human-friendly shorthand for binary:

The nibble link

One hex digit represents exactly **one nibble** (4 bits), because both have 16 possible values. Two hex digits represent exactly **one byte**. That is the entire reason hexadecimal is used in computing.

Denary	Binary	Hex	Denary	Binary	Hex
0	0000	0	8	1000	8
1	0001	1	9	1001	9
2	0010	2	10	1010	A
3	0011	3	11	1011	B
4	0100	4	12	1100	C
5	0101	5	13	1101	D
6	0110	6	14	1110	E
7	0111	7	15	1111	F

You have already met hex in the wild: colour codes like #FF6600 are three bytes — red, green and blue — each written as two hex digits.

Practice 9.1 — Fill in the missing values:

- a) denary 11 = hex **B** b) hex E = denary **14** c) binary 0111 = hex **7**
d) hex C = binary **1100** e) denary 15 = hex **F** f) binary 1010 = hex **A**

Hex place value

Hexadecimal follows the same place-value rule as denary and binary — but each column is worth **sixteen times** the one before:

Four-thousand-and-ninety-sixes	Two-hundred-and-fifty-sixes	Sixteens	Units
16^3	16^2	16^1	16^0
4,096	256	16	1

In the exam you only meet **two-digit** hex numbers (up to FF), so the columns you will actually use are the **sixteens** and the **units**.

Worked example

Convert **8B** to denary.

	Sixteens	Units
Hex digit	8	B
As denary	8	11

$(8 \times 16) + (11 \times 1) = 128 + 11 = \mathbf{139}$

Maths Link – your 16 times table

You already know it — it is the 8 times table, doubled: $4 \times 8 = 32$ so $4 \times 16 = 64$.

1×16	2×16	3×16	4×16	5×16	6×16	7×16	8×16
16	32	48	64	80	96	112	128
9×16	10×16	11×16	12×16	13×16	14×16	15×16	16×16
144	160	176	192	208	224	240	256

Practice 10.1 — Use the sixteens-and-units columns to convert:

a) $4F = \underline{\mathbf{79}}$ b) $26 = \underline{\mathbf{38}}$ c) $3C = \underline{\mathbf{60}}$

a) $4 \times 16 + 15 = 79$ **b)** $2 \times 16 + 6 = 38$ **c)** $3 \times 16 + 12 = 60$

Exam question [3 marks]

OCR J277/01 · June 2024 · Q1(d)

Describe how to convert a 2-digit hexadecimal number into denary. Use an example in your answer.

Either — multiply the left/first digit by 16, then add the value of the second digit. **Or** — convert each digit into 4-bit binary, combine, and convert the 8-bit binary to denary.

e.g. $2F = 2 \times 16 + 15 = 47$.

1 mark for a correct worked example; 1 mark each to max 2 for the stages. No marks for converting denary → hex.

Binary ↔ hexadecimal — the nibble trick

Because one hex digit *is* one nibble, converting between binary and hex never needs any arithmetic bigger than $8 + 4 + 2 + 1$.

Binary → hex: split the bit pattern into nibbles (from the right — add 0s to the front if needed), convert each nibble to denary with 8–4–2–1, then write each as a hex digit.

Worked example

Convert 1101 1001 to hexadecimal.

First nibble: $1101 = 8 + 4 + 1 = 13 = \mathbf{D}$ Second nibble: $1001 = 8 + 1 = 9 = \mathbf{9}$

1101 1001 = **D9**

Hex → binary: the same trick in reverse — turn each hex digit into its 4-bit nibble.

Worked example

Convert **F2** to binary.

$F = 15 = 8 + 4 + 2 + 1 = 1111$ $2 = 0 + 0 + 2 + 0 = 0010$

F2 = 1111 0010

Practice 11.1 — Binary to hexadecimal:

a) 1001 = **9** b) 1111 = **F** c) 0110 1110 = **6E** d) 0101 0111 = **57**

Practice 11.2 — Hexadecimal to binary:

a) 5C = **0101 1100** b) 3E = **0011 1110** c) A4 = **1010 0100** d) E7 = **1110 0111**

Practice 11.3 — A pixel's red value is stored as the byte 1010 0010. Write it as two hex digits.
A2

Denary → hexadecimal

Divide by powers of 16, largest first, keeping whole numbers only. Each answer is one hex digit; each remainder feeds the next step.

Worked example

Convert **495** to hexadecimal.

Step 1 — the largest power of 16 below 495 is 256.

$495 \div 256 = 1$ (ignore the decimals) → first digit **1**

remainder: $495 - (1 \times 256) = 239$

Step 2 — divide the remainder by 16.

$239 \div 16 = 14$ → second digit $14 = \mathbf{E}$

remainder: $239 - (14 \times 16) = 239 - 224 = 15$

Step 3 — divide the remainder by 1.

$15 \div 1 = 15$ → third digit $15 = \mathbf{F}$

$495 = \mathbf{1EF}$

For exam-sized numbers (up to 255) there is only one division:

Worked example

Convert **235** to hexadecimal.

$235 \div 16 = 14$ remainder 11 $14 = \mathbf{E}$ and $11 = \mathbf{B}$ so $235 = \mathbf{EB}$

Practice 12.1 — Convert to hexadecimal. Show your divisions.

Denary	Working	Hex
a) 35	$35 \div 16 = 2 \mathbf{r} 3$	<u>23</u>
b) 176	$176 \div 16 = 11 \mathbf{r} 0$	<u>B0</u>
c) 201	$201 \div 16 = 12 \mathbf{r} 9$	<u>C9</u>
d) 255	$255 \div 16 = 15 \mathbf{r} 15$	<u>FF</u>

Stretch — beyond the exam limit: convert 300 to hex. **12C** $300 \div 256 = 1 \mathbf{r} 44$; $44 \div 16 = 2 \mathbf{r} 12$; $12 = \mathbf{C}$.

Hexadecimal → denary

Multiply each digit by its column value and add. Letters go through their denary values first: A = 10, B = 11, C = 12, D = 13, E = 14, F = 15.

Worked example

Convert **B7** to denary.

$$\begin{array}{r} \text{units digit: } 7 \times 16^0 = 7 \times 1 = 7 \\ \text{sixteens digit: } B \times 16^1 = 11 \times 16 = 176 \\ \hline \text{total: } 176 + 7 = \mathbf{183} \end{array}$$

Check both ways

Reverse it: $183 \div 16 = 11$ remainder 7, and $11 = B$. ✓

Practice 13.1 — Convert to denary. Show your working.

Hex	Working	Denary
a) D	13×1	<u>13</u>
b) 3C	$3 \times 16 + 12$	<u>60</u>
c) 5D	$5 \times 16 + 13$	<u>93</u>
d) E2	$14 \times 16 + 2$	<u>226</u>
e) F3	$15 \times 16 + 3$	<u>243</u>

Stretch — convert 1EF back to denary. Do you get 495?

$$1 \times 256 + 14 \times 16 + 15 \times 1 = 256 + 224 + 15 = 495. \checkmark$$

Mixed practice

No calculators — just like the exam. Write the binary line before you start.

Exam question [3 marks]

Convert the following binary numbers to denary:

a) 1111 = 15 b) 0101 0101 = 85 c) 1010 1010 = 170

Exam question [3 marks]

Convert the following denary numbers to binary:

a) 7 = 0000 0111 b) 35 = 0010 0011 c) 182 = 1011 0110

Exam question [2 marks]

Convert the following hexadecimal numbers to denary:

a) C = 12 b) 5D = 93

Exam question [2 marks]

Convert the following denary numbers to hexadecimal:

a) 47 = 2F b) 199 = C7

Exam question [2 marks]

a) Convert 0111 1101 to hexadecimal: 7D

b) Convert 4F to binary: 0100 1111

Exam question [2 marks]

Explain why computers store data in binary.

**Computers are built from switches with two states, on/off (1 mark);
two states can be represented and read reliably (1 mark).**

Exam question [2 marks]

Explain why programmers often use hexadecimal to represent binary values.

**Shorter / easier for humans to read, write and remember (1 mark);
each hex digit maps exactly to one nibble, so conversion is simple (1 mark).**

Exam question [4 marks]

OCR J277/01 · June 2023 · Q1(b)

Complete the table by writing the missing denary, 8-bit binary or hexadecimal values.

Denary	8-bit binary	Hexadecimal
<u>7</u>	00000111	7
49	<u>00110001</u>	31
<u>102</u>	01100110	66
244	11110100	<u>F4</u>

Adding binary numbers

You can add two binary numbers straight in binary, without converting to denary first. It works exactly like column addition in denary — start at the right, carry to the left — but there are only **four rules** to learn:

Key concept

$$0 + 0 = 0 \quad 0 + 1 = 1 \quad 1 + 1 = 10 \quad 1 + 1 + 1 = 11$$

The last two are the only ones that carry. $1 + 1$ writes **0** and carries **1**; $1 + 1 + 1$ writes **1** and carries **1**.

Worked example

Add 0011 0101 and 1000 0010. Work right to left, writing any carry above the next column.

$$\begin{array}{r} 00110101 \\ + 10000010 \\ \hline 10110111 \end{array}$$

Check: $53 + 130 = 183$, and 1011 0111 is 183. ✓

Practice 15.1 — Add these binary numbers. Show your carries.

$$\begin{array}{r} 0011 \\ + 0100 \\ \hline 0111 \end{array}$$

$$\begin{array}{r} 0001 \\ + 1000 \\ \hline 1001 \end{array}$$

$$\begin{array}{r} 11010010 \\ + 00101101 \\ \hline 11111111 \end{array}$$

Overflow

A byte holds eight bits, and the largest value eight bits can hold is **255**. If an addition needs a ninth bit, that bit has nowhere to go — it is simply lost. This is an **overflow error**.

Worked example

Add 0111 0001 and 1001 1110. $113 + 158 = 271$.

$$\begin{array}{r} 01110001 \\ + 10011110 \\ \hline 100001111 \end{array}$$

The answer needs **nine** bits. Only eight are stored, so the computer keeps 0000 1111 — which is 15, not 271. The bold ninth bit has fallen off the end. **That is an overflow error.**

Exam question [2 marks]

OCR J277/01 · June 2024 · Q1(e)

Add these two 8-bit binary numbers using binary addition. Show your working out.

$$\begin{array}{r} 01101011 \\ + 00001111 \\ \hline 01111010 \end{array}$$

Binary shifts

Sliding every bit one place to the **left** doubles the number. Sliding every bit one place to the **right** halves it. That is all a **binary shift** is — and it is how computers multiply and divide quickly.

Key concept

Left shift multiplies by 2 for every place moved

Right shift divides by 2 for every place moved

So 1 place = $\times 2$, 2 places = $\times 4$, 3 places = $\times 8$, 4 places = $\times 16$. Gaps left behind are filled with 0.

Worked example

$0001\ 0010 \times 8$. Multiplying by 8 is a **left shift of 3 places**.

$0001\ 0010 \rightarrow 1001\ 0000$

Check: $18 \times 8 = 144$, and $1001\ 0000$ is 144. ✓

Practice 16.1 — Use a left shift. Give each answer in 8 bits.

a) $0101 \times 2 =$ 0000 1010

b) $1100 \times 8 =$ 0110 0000

c) $0101\ 1110 \times 2 =$ 1011 1100

Practice 16.2 — Use a right shift. Give each answer in 8 bits.

a) $0110 \div 2 =$ 0000 0011

b) $0011\ 1000 \div 8 =$ 0000 0111

c) $0101\ 1110 \div 2 =$ 0010 1111

Key concept

What falls off the end is gone. Shift $0111\ 1101$ (125) left by 2 to multiply by 4 and you need 500 — far past 255, so bits drop off the left and the answer is wrong: that is **overflow** again. Shift $0001\ 0011$ (19) right by 1 to halve it and you get $0000\ 1001$ (9), not 9.5 — the bit that fell off the right was the half, and binary shifts cannot keep it. **Shifting loses information at both ends.**

Practice 16.3 — $1011\ 1011 \times 8$. Explain what goes wrong.

Exam question [2 marks]

OCR J277/01 · June 2025 · Q5(e)

Describe the binary shift that can be used to multiply any number by 8.

left shift • 3 places

Reference — the cheat sheet

The three place-value lines. Same rule every time: multiply each digit by its column, add them up.

Binary (base 2)	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
	128	64	32	16	8	4	2	1
Denary (base 10)					10^3	10^2	10^1	10^0
					1,000	100	10	1
Hex (base 16)					16^3	16^2	16^1	16^0
					4,096	256	16	1

The sixteen hex digits and their nibbles.

Denary	Binary	Hex	Denary	Binary	Hex
0	0000	0	8	1000	8
1	0001	1	9	1001	9
2	0010	2	10	1010	A
3	0011	3	11	1011	B
4	0100	4	12	1100	C
5	0101	5	13	1101	D
6	0110	6	14	1110	E
7	0111	7	15	1111	F

Units of storage. *Kids Must Go To Parties* — kB, MB, GB, TB, PB.

8 bits	= 1 byte	1,000 GB	= 1 TB
4 bits	= 1 nibble	1,000 MB	= 1 GB
1,000 bytes	= 1 kB	1,000 kB	= 1 MB
1,000 TB	= 1 PB	1 byte	= 256 values (0–255)

The four methods.

- **binary** → **denary**: binary line, add the 1-columns
- **denary** → **binary**: divide by 2, read remainders bottom-up
- **binary** ↔ **hex**: one nibble = one hex digit
- **denary** ↔ **hex**: multiply by 16s, or divide by 16 and keep the remainder

bit	a single binary digit, 1 or 0 — the smallest unit of data
bit pattern	a group of bits joined together to represent data or instructions
nibble	4 bits — half a byte; holds 16 values, and matches one hex digit
byte	8 bits — the smallest bit pattern normally used to represent data; holds 256 values
number base	how many different digits a number system uses
denary	base 10 — digits 0–9; also called decimal
binary	base 2 — digits 0 and 1; the form all computer data takes
hexadecimal	base 16 — digits 0–9 and A–F; shorthand for binary, one digit per nibble
SI units	the international system of measurement prefixes: kilo, mega, giga, tera, peta
kilobyte (kB)	1,000 bytes (10^3); then MB, GB, TB, PB — each 1,000 times the last

Can I...?

Tick a column for each skill. Be honest — the point is to find what to practise before the end-of-unit test, not to collect ticks.

	Not yet	Getting there	Confident
Explain why data must be stored in binary			
Name the units of storage in order: bit, nibble, byte, kB, MB, GB, TB, PB			
Calculate data capacity and storage requirements			
Convert binary → denary (add the 1-columns)			
Convert denary → binary (divide by 2, remainders bottom-up)			
Convert binary ↔ hex (one nibble = one hex digit)			
Convert denary ↔ hex (multiply by 16s, or divide by 16)			
Add two 8-bit binary numbers, and say what overflow means			
Perform a binary shift and say what it does to the value			

The one thing I most need to practise:

Notes